

B.E. Even Semester Midterm Examination, 2024-25

Subject: Discrete Mathematics

Paper Code: PCC-IT401

Full Marks: 15 | **Time:** 1 Hr.

Q1. Define a bipartite graph. Draw .

Answer:

A **bipartite graph** is a graph whose vertices can be divided into two disjoint and independent sets and such that every edge connects a vertex in to one in . There are no edges connecting vertices within the same set.

Drawing (Complete Bipartite Graph):

has two sets of vertices, and . Every vertex in is connected to every vertex in .

u_1 ----- v_1

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\ /

u_2 -X- v_2

/ \

/ \

u_3 ----- v_3

(Each u connects to v_1 , v_2 , and v_3)

Q2. Show that forms a group but does not form a group.

Answer:

1. is a group:

A set forms a group under an operation if it satisfies four properties:

- **Closure:** For any , their sum is also in .
- **Associativity:** For any , .
- **Identity:** There exists an element such that for all .
- **Inverse:** For every , there exists an inverse such that .

Since all 4 properties hold, is a group.

2. is NOT a group:

While real numbers under multiplication satisfy Closure, Associativity, and Identity (where 1 is the identity element), it fails the **Inverse property**.

For an element , its multiplicative inverse is . However, for the element , the inverse is undefined. Since has no inverse, does not form a group.

Q3. Define a group with an example. Give an example of a monoid which is not a group with reason.

Answer:

Definition of a Group:

A group is a set combined with a binary operation that satisfies four axioms:

- **Closure:**
- **Associativity:**
- **Identity:** such that
- **Inverse:** such that

Example: The set of integers under addition is a group.

Monoid that is not a group:

A **monoid** satisfies Closure, Associativity, and Identity, but lacks the requirement for Inverses.

Example: The set of Natural numbers including zero under addition where is a monoid. The identity is . It is **not a group** because elements (like) do not have an additive inverse () within the set of Natural numbers.

Q4. Draw a complete graph with five vertices. Verify Handshaking theorem for the graph.

Answer:

Complete Graph :

A complete graph has 5 vertices where every vertex is connected to every other vertex exactly once.

- Number of vertices () = 5
- Number of edges () =
- Degree of each vertex =

Handshaking Theorem Verification:

The Handshaking Theorem states that the sum of the degrees of all vertices in a graph is equal to twice the number of edges:

- **LHS (Sum of degrees):** 5 vertices degree 4 = 20
- **RHS ():**

Since LHS = RHS (), the theorem is verified.

Q5. Draw a graph G with six vertices. Find four subgraphs of this graph G with at least four vertices.

Answer:

Let graph be a simple cycle graph defined by vertices and edges forming a closed loop: .

Four subgraphs with at least 4 vertices:

- **Subgraph 1 (Path of 5 vertices):** Remove vertex 6 and its edges.
Vertices: , Edges: .
- **Subgraph 2 (Path of 4 vertices):** Remove vertices 5 and 6.
Vertices: , Edges: .
- **Subgraph 3 (Disconnected 4 vertices):** Remove vertices 1 and 4.
Vertices: , Edges: .
- **Subgraph 4 (Graph G itself without one edge):** Vertices: , Edges: . (This is a 6-vertex spanning subgraph).